

9.camera

```
from picamera import PiCamera
from time import sleep
```

```
camera = PiCamera()
camera.rotation = 180
```

```
camera.start_preview()
sleep(5)
camera.capture('/home/pi/Desktop/image.jpg')
camera.stop_preview()
```

```
from picamera import PiCamera, Color
from time import sleep
```

```
camera = PiCamera()
camera.rotation = 180
```

```
camera.start_preview()
```

```
camera.annotate_background = Color('yellow')
camera.annotate_foreground = Color('blue')
camera.annotate_text_size = 80
camera.annotate_text = "Hello KMIT"
```

```
sleep(5)
camera.capture('/home/pi/Desktop/modified.jpg')
camera.stop_preview()
```

```
from picamera import PiCamera, Color
from time import sleep
```

```
camera = PiCamera()
camera.rotation = 180
```

```
camera.start_preview()
```

```
camera.annotate_background = Color('yellow')
camera.annotate_foreground = Color('blue')
camera.annotate_text_size = 80
```

```
camera.annotate_text = "KMIT IT"

camera.start_recording('/home/pi/Desktop/video.h264')
sleep(5)
camera.stop_recording()

camera.stop_preview()
```

8.ultra

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BCM)

GPIO_TRIGGER = 18
GPIO_ECHO = 24

GPIO.setup(GPIO_TRIGGER, GPIO.OUT)
GPIO.setup(GPIO_ECHO, GPIO.IN)

def distance():
    GPIO.output(GPIO_TRIGGER, True)
    time.sleep(0.00001)
    GPIO.output(GPIO_TRIGGER, False)

    start = time.time()
    stop = time.time()

    while GPIO.input(GPIO_ECHO) == 0:
        start = time.time()

    while GPIO.input(GPIO_ECHO) == 1:
        stop = time.time()

    time_elapsed = stop - start
    dist = (time_elapsed * 34300) / 2

    return dist
```

```
while True:
    d = distance()
    print("Distance:", d, "cm")
    time.sleep(1)
```

6.pir

```
import RPi.GPIO as GPIO
import time

GPIO.setwarnings(False)
GPIO.setmode(GPIO.BOARD)

# PIR input pin
GPIO.setup(11, GPIO.IN)

# LED output pin
GPIO.setup(3, GPIO.OUT)

while True:
    if GPIO.input(11):
        print("Motion Detected!")
        GPIO.output(3, True)
        time.sleep(1)
    else:
        print("No Motion")
        GPIO.output(3, False)
        time.sleep(1)
```

6.led

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BOARD)

while True:
    GPIO.setup(8, GPIO.OUT)
```

```
GPIO.output(8, GPIO.HIGH)  
time.sleep(2)
```

```
GPIO.output(8, GPIO.LOW)  
time.sleep(2)
```